KEVIN M. FEDORUK

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PROFESSIONAL SUMMARY

Mechanical engineer seeking an entry-level design, propulsion, or test engineering role in aerospace and defense. B.S. in Mechanical Engineering (Propulsion concentration) from Embry-Riddle Aeronautical University and an MBA from Monmouth University, with hands-on experience across the full design cycle, including structural and thermal analysis, FEA, CAD modeling with GD&T, and instrumented testing on fabricated hardware. U.S. citizen able to obtain a Secret clearance. Proven leadership and performance under pressure through six years of NCAA Division I / NAIA collegiate athletics.

TECHNICAL SKILLS

Software: SOLIDWORKS, MATLAB, ANSYS Workbench, LabVIEW, Microsoft Excel/Office
Engineering: Structural Analysis (Free-Body Diagrams, Hand Calculations), Fatigue & Damage Tolerance (DE-Goodman), Finite Element Analysis, CAD Modeling & GD&T, Mechanical & Machine Design, Mechanism/Actuation Design, AGMA Gear Stress Analysis, Shaft & Bearing Selection, Propulsion Systems (Turbojet, Turbofan, Ramjet, Rocket), Thermodynamics, Fluid Mechanics
Manufacturing & Testing: CNC Machining, MIG/TIG Welding, 3D Printing, Pressure Transducers, Thermocouples, Strain Gages
Languages: English, Spanish (Seal of Biliteracy)

ENGINEERING PROJECT EXPERIENCE

Variable-Area Exhaust Nozzle — Personal SolidWorks Project *Sep. 2026 – Present*

- Designing a 12-primary/12-secondary-petal variable-area exhaust-nozzle mechanism in SolidWorks, featuring a guided axial actuation ring, synchronized linkages, and a dual-servo actuation concept.
- Iteratively redesigned the primary petal hinge/crank geometry to reduce mechanism packaging while preserving actuation kinematics, and verified in CAD a 3.85 mm actuator stroke with interference-free operation from 0° to 15.5° petal rotation.
- Preparing production drawings with GD&T, which are nearly complete, and 3D printing the full assembly to test fit, kinematics, and servo actuation.

Senior Capstone – Rapid Lunar Transport Vehicle (RLTV) | Team Lead, 6-Person Team *Aug. 2023 – May 2024*

- Led a six-person team ("LUNA EV") through a NASA Artemis-aligned capstone; after helping design the full two-wheeled RLTV in preliminary design, focused detail design on a thermal protection system for its 250 W electric hub motors.
- Selected a 44°C-melt paraffin wax over a 54.67°C alternative through empirical testing, sized it via first-principles heat-transfer calculations (0.915 kg / 1.10 L required vs. 2.10 L designed capacity), and validated melt behavior with an ANSYS transient thermal simulation.
- Designed, fabricated, and instrumented a custom test article and rig, with a TIG-welded aluminum shell, brazed 16-compartment wax housing, dual-probe temperature logging (including a thermocouple in the motor core), and a digital torque sensor, to validate against a 30-minute, -10°C to 50°C requirement.
- Measured 73.4 W of real motor heat dissipation during testing; the system met the required 30-minute limit, and extended testing informed a recommendation for a lower-melting-point wax to add more margin, delivering the project $565.89 under budget.

Multi-Stage Gearbox — Machine Design Project | 3-Person Team *Aug. 2023 – Dec. 2023*

- Designed a four-gear, three-shaft gearbox to lift a 50-lb load over a required 10^8-cycle life, performing full AGMA gear stress analysis and DE-Goodman shaft fatigue analysis by hand, with every gear and shaft clearing its factor of safety (min. 30.97 bending, 3.62 contact, 3.78 shaft).
- Modeled the complete gear train and housing in SOLIDWORKS with GD&T, sourcing hardware and evaluating manufacturability and cost tradeoffs to finalize a design that beat both the size and $400 budget constraints, delivered at $331.31.

MBA Marketplace Business Strategy Simulation | Team President, 9-Week Simulation *Jan. 2026 – May 2026*

- Directed a nine-week global business simulation, managing product development, pricing, production planning, advertising, and market expansion across five customer segments, three geographic regions, and multiple metropolitan markets.
- Finished 1st place in class and top 10% worldwide among more than 1 million Marketplace simulation participants.

EDUCATION

Embry-Riddle Aeronautical University — B.S., Mechanical Engineering, Propulsion *Aug. 2020 – May 2024*
GPA 3.602/4.000, Cum Laude. Relevant Coursework: Dynamics, Solid Mechanics, Machine Design, Computer-Aided Design, Heat Transfer, Thermodynamics, Fluid Mechanics, Airbreathing & Rocket Propulsion, Advanced Space Propulsion, Machine Shop, Electrical Engineering

Monmouth University — Master of Business Administration *Sep. 2025 – Aug. 2026*
GPA 3.910/4.000. Relevant Coursework: Managerial Economics, Managing & Leading, Project Management, Strategic Management

University of California, Riverside — MBA Coursework Completed *Sep. 2024 – Jun. 2025*

LEADERSHIP EXPERIENCE

MBA Marketplace Simulation Team President — Monmouth University *Jan. 2026 – May 2026*
Senior Capstone Team Lead — Embry-Riddle Aeronautical University *Aug. 2023 – May 2024*
Student-Athlete Advisory Committee (SAAC) Representative — Embry-Riddle Aeronautical University *Aug. 2020 – May 2024*

COLLEGIATE ATHLETICS

Six years of NCAA Division I / NAIA collegiate baseball — Monmouth University (2025–2026), University of California, Riverside (2024–2025), Embry-Riddle Aeronautical University (2020–2024) — while completing rigorous engineering and MBA coursework, developing leadership, communication, time management, resilience, teamwork, and the ability to perform under pressure.

AWARDS & SCHOLARSHIPS

Order of the Engineer (Inducted Member, May 2024) • Monmouth University Full Athletic Scholarship (2025–2026) • University of California, Riverside Academic Scholarship (2024–2025) • Embry-Riddle Aeronautical University Presidential Scholarship (2020–2024) • Embry-Riddle Aeronautical University Athletic Scholarship (2020–2024) • Embry-Riddle Aeronautical University Dean's List (2020–2024)